Blessed G/Michael Catholic Secondary School

1st Semester Biology final examination for Grade 11

January, 2018 E.C Time Allowed -30min'

Name Alubamin Menota Grade 11 Section 11C No 33

I. Write true if the statement is correct or false if the statement is incorrect.

True 1. Pepsin hydrolyses peptide bonds next to aromatic amino acids only.

True 2. The turn over number of molecules is the number of substrates converted by one enzyme molecule per second at saturated (fully occupied) active sites.

false 3. The importance of adding hope flower into the solution during beer production is to add sweet taste and better flavor.

True 4. Deamination is the anabolic process which occurs on amino acids to release the amino group and CO_2.

True 5. Metabolites are chemical signals that affect the behavior of other glands and tissues.

True 6. The change in the behavior of humans by life experiences is learning.

II. Choose the best answer from the given alternatives and write your answers in Capital letter.

D 7. Which property of DNA makes it different from RNA? It is:-

A. Double stranded B. Double helix C. Self-replicating D. All of them.

D 8. When heterozygous yellow round seed plants are let to self-fertilize, the frequency of occurrence of RrYY genotype among the offspring's is Yy

A. 1/16 B. 3/16 C. 4/16 D. 2/16

C 9. From the self-crossing of F1 = TtRr × TtRr, what is the probability of getting dwarf wrinkled F2 generation (hint T= tall, t= dwarf, R= round, r= wrinkled).

A. 1/8. B. 3/16 C.1/16 D. 1/4

A 10. Which one is not the unique future of meiosis:-

A. Identical daughter cells formation

B. Synapsis formation

C. Reduction cell division

D. Crossing over occurrence

A 11. Which component of transcribed RNA in eukaryotes is present in the initial transcript but is removed before translation occurs.

A. Intron B. 5'-G cap C. Protein coding exons D. Ribosome binding site

D 12. Individuals having dissimilar traits (alleles) on homologous chromosomes are called

A. Recessive B. Homozygous C. Dominant D. Heterozygous

C ⑬ A man with type O blood married a woman with heterozygous type A blood. What are the possible phenotypes of the children?

A. 25% chance of type 'A'

B. 25% chance of type 'B'

C. 50% chance of type 'O'

D. 75% chance of type 'A'

O O

O OO OO

A OA OA

D 14. Which one of the following is not the significance of genetics?

A. Production of Antibiotics through genetic engineering

B. It has roles in the improvement of agriculture

C. DNA finger printing for suspected criminal identification which helps to formulate law

D. It has roles on the studying plant classifications

Bb b Bb + bb
b Bb bb
b Bb bb

D 15. When dominant *Bb* and recessive *bb* is crossed, the percentage of progeny showing the parental genotype is

A. 25% B. 75% C. 50% D. 100%

A 16. Given the sequence on the coding strand of DNA is AGCATGAC then what is strand of mRNA transcribed?
UCGUACUG

A. UCGUACUG B. UCGTACAG C. AGCAUGAC D. TCGTACTG

B 17. Alleles are paired in____and separated in_____.

A. Ovum, Zygote. C. Dihybrid, Ovum.

B. Diploid organism. Ovules. D. Polyploidy, Pollen.

A 18. Which one of the following has no roll in protein synthesis?

A. DNA Polymerase. B. DNA Helicase. C. RNA Polymerase. D. RNA Primase.

C 19. In which type of inhibition does the inhibitor attaches only on the enzyme substrate complex?

A. Competitive inhibitor. C. Uncompetitive inhibitor.

B. Non-competitive inhibitor. D. Irreversible

A 20. Hibernation is a behavior that allows animals to:

A. Adapt to changes in temperature

B. Communicate with each other

C. Defend their territory

D. Conserve energy during periods of food scarcity in cold seasons.

B 21. Why is an ectotherm /Poikilotherm/ likely to survive longer without food than an endotherm of equal size

A. The ectotherm has a higher metabolic rate

B. The ectotherm does not use energy to regulate its temperature ✓

C. The ectotherm expends more energy than the endotherm

D. The ectotherm is better able to metabolize its stored energy

B ㉒ The RNA and protein synthesis occurs in______

A. M phase B. S phase C. G1 Phase D. G2 phase

C 23. During meiosis, chromatids of individual chromosomes separate at ___?

A. Metaphase I B. Anaphase I C. Anaphase II D. Metaphase II

B ㉔ Spindle fibers are formed by___________

A. Spindles B. Tubulin C. Flagella D. Globulin

C 25. Which of the following enzyme inhibitions shows decreased Km Value ?

A. Competitive inhibition C. Un-competitive inhibition

B. Non-competitive inhibition D. Feed back inhibition

increase - compet
decrease - un
Constant - non

B 26. The enzymes which break up starch into sugar are called

A. Hydrolases B. Amylases C. Lipases D. Nucleases

A 27. Which one is a true composition of Apoenzyme and Coenzyme respectively

A. Protein and Vitamin C. Amino acid and an inorganic compound

B. Vitamin and inorganic compound D. Protein and metal E. A& C

D 28. In competitive inhibition, inhibitors bears a close structural similarity with the

A. Co-enzyme B. Co-factor C. Prosthetic group D. Substrate

B 29. Enzyme acts best at a particular temperature called

A. Catalytic Temperature C. At normal Body temperature

B. Optimum temperature D. None of the above

D 30. Which bond is not associated with Enzyme-substrate interaction?

A. Hydrogen bonds
B. Di-sulfide bonds ←Covalent
C. Ionic bonds
D. Van deer Waal's force of attraction

D 31. If body temperature decreases, which one of the following is incorrect?

A. Vasoconstriction
B. Increase metabolic rate
C. Hair raised
D. Sweating

C 32. Homeostasis is best described by which of the following statements?

A. Maintaining a stable and unaltered state of the body.
B. Stable equilibrium
C. Keeping the internal environment near-constant.
D. Changing the external environment to meet the body's requirements

C 33. The organization of group of animals in such a way that some members of the group have greater access to resources like food or mates than others is called:

A. Agnostic
B. Territory
C. Hierarchy
D. Altruism

B 34. Which type of learning happens in a narrow window of time shortly after birth?

A. Associative learning
B. Imprinting
C. Habituation
D. Classical conditioning

35. Which of the following layer develops first during embryonic development?

A. Ectoderm
B. Endoderm
C. Mesoderm
D. Both B and C

36. Competitive inhibition includes all of the following except

A. Chemical structure of the inhibitor closely resembles that of the substrate.
B. Combines reversibly with the enzyme
C. When both the substrate and the inhibitor are present they compete for the same binding site.
D.V max is decreased
E.Is removed by increases the concentration of inhibitor

37. Reversible noncompetitive Inhibition includes all of the following except:-

A. I combine with the enzyme away from the catalytic site.
B. The I is not similar to the substrate in structure.
C. The I can combine with the free enzyme or with the enzyme substrate
D. Increase the Km

D 38. The function of placenta is :-

A. transports oxygen from the mother to the embryo.
B. removes waste from the embryo's environment
C. It protects the embryo during pregnancy from physical damage.
D. All except C

III. Write Short answers.

39. Write the sequential steps of malting

A. Steeping B. Germination C. Kilning D. Fermentation

43. The sleep-wake cycles in our body is controlled the hormone [illegible]

[illegible]

RrYy RrYy

	RY	Ry	rY	ry
RY	RRYY	RRYy	[illegible]	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

RrRry
RY Ry rY ry
RY RRYY
~~Ry~~ RRYy
rY RrYY
ry RrYy

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Time Allowed -50min'

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January, 2018 E.C

Name Aluhami Menole Grade 11 Section 11C No 33

Write the Answers in capital letters

True / False	M. Choice	14. D [redacted]	22 [redacted] B	30 [redacted] D B	38 [redacted] D
1. True	7. D .	15. D [redacted]	23 [redacted] C	31 [redacted] D	Short answer
2. True	8. D ..	16. A [redacted]	24 [redacted] B A	32 [redacted] C	39. A. [redacted] Stepping
3. False	9. C	17 [redacted] B	25 [redacted] C	33 [redacted] C	B. [redacted] Germinating
4. True	10. A .	18. A [redacted]	26 [redacted] A	34 [redacted] A	C [redacted] Kilning
5. True	11. A .	19. C [redacted]	27 [redacted] A	35 [redacted] A	D [redacted] fermentation
6. True	12. D ..	20. A [redacted]	28 [redacted] D	36 [redacted] C	40. [redacted] ~~[illegible]~~
	13. C .	21. A [redacted]	29 [redacted] B	37 [redacted] D	~~[illegible]~~

~~[illegible]~~

40. Suberoral [illegible]cadan